Critical Design Review

Structures

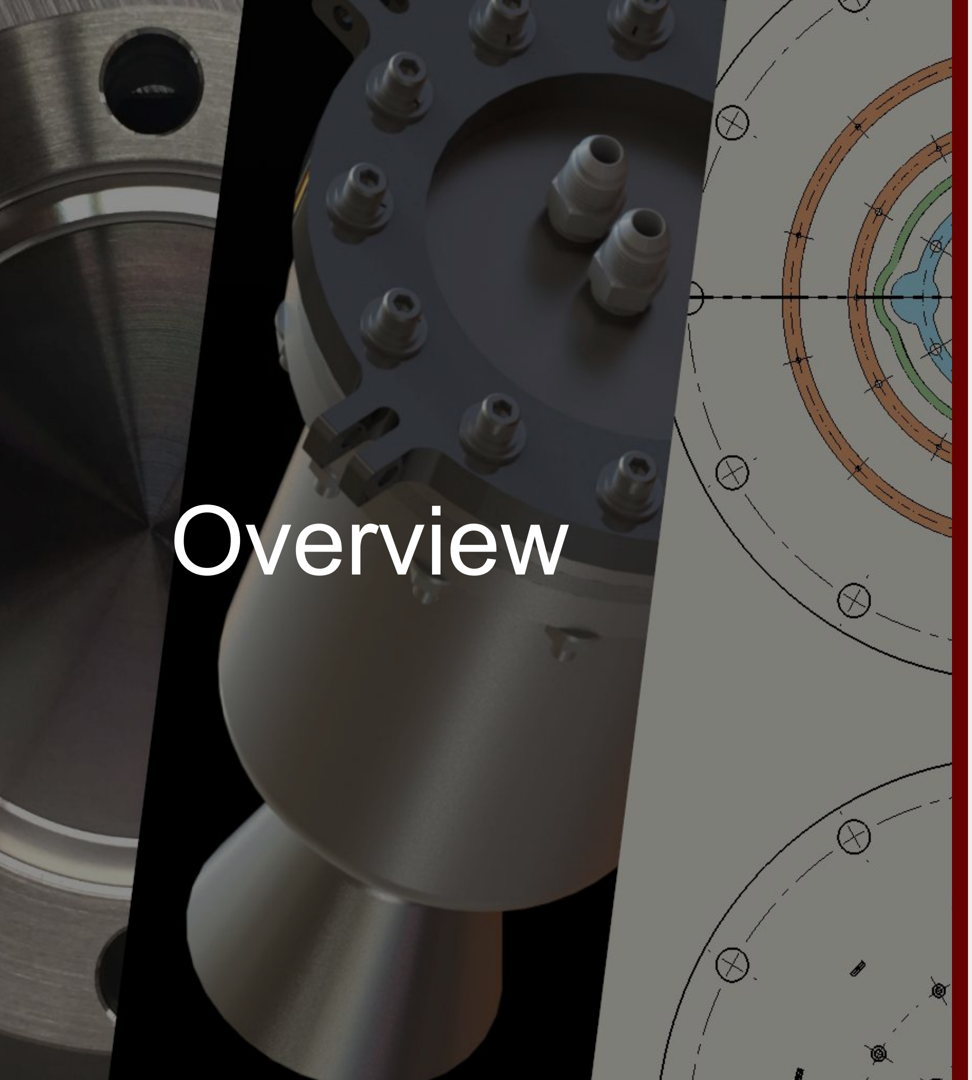
(1 Hour Estimate)

Structures System Requirements

#	Requirement	Rationale	Verification
1	Test stand assembly (all structural components and subcomponents) shall maintain a FOS of **4** (yield strength).	Upper end of expected thrust curve is 1500 lbf upwards. A safety factor of 4 ensures that even under unexpected dynamic loads—such as vibration, wind gusts, or misfires—the test stand remains structurally sound.	FEM, Test
2	Test stand will withstand 730 lbf horizontally 4*1500*sin(7 degrees)	Test stand must be prepared for loads in any direction for additional safety	FEM, Test
3	Test stand will not deform over time	Test stand should be made to withstand gimbaling and thrust vectoring	Vibration Testing, FEM

Structures System Requirements

#	Requirement	Rationale	Verification
4	Test stand will not move more than 0.5 in from its initial position	To ensure accurate readings, the stand should not move from its original position	Design Analysis, Test
5	Flame diverter can handle heat from the engine, 1500K for 15s, without damage to critical components	To protect critical components and allow for reuse. The test stand must divert the flame away from components.	Controlled Burn Testing
6	The engine will be able to move 5-7 degrees in every direction while engine is firing	In order accurately control attitude and flight path in flight, the engine must be able to be gimbaled	Dry Run Test



Overview

(Time Estimate)

I. Aluminum Stand
II. Gimbal
III. Wooden Stand
IV. Flame Diverter
V. Bottle Stand
VI. Full Assembly

Aluminum Frame

- Penetrators
- Horseshoes
- Extrusions
- 3’ plate

Structural Requirements

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3	Test stand will not deform over time	Test stand should be made to withstand gimbaling and thrust vectoring	Vibration Testing, FEM
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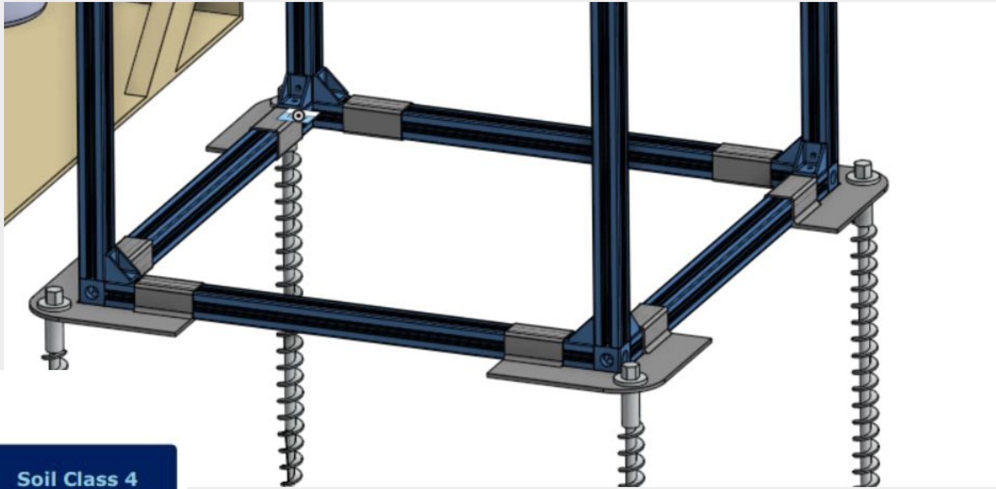
Requirement #1: Axial Load Integrity

- The rated pullout force for our penetrator style anchors for class 2 soil is 3,100 lbs
- 4 Corner Penetrators
- Rated to 12,400 lbs in total



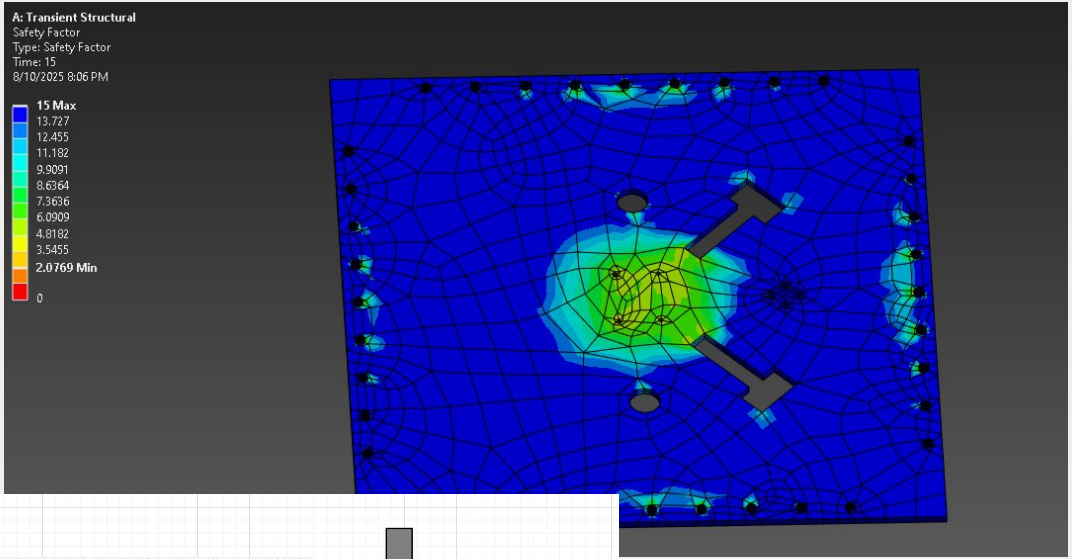
LOAD CAPACITY

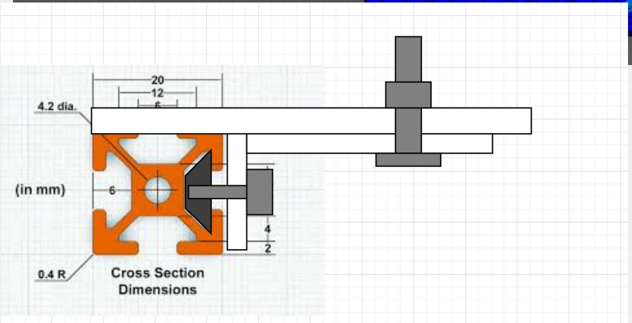
Pullout strength with flight fully embedded

Soil Class 1 Hardpan Asphalt	Soil Class 2 Sandy gravel Very dense sand	Soil Class 3 Silty/clayey sand Silty gravel	Soil Class 4 Loose/med dense sands Loose sands Firm clays	Soil Class 4 Loose fine un-compacted sand
4,500 lb 20.0 kN	3,100 lb 13.8 kN	1,100 lb 4.89 kN	630 lb 2.80 kN	360 lb 1.60 kN

Requirement #1: Axial Load Integrity

- ~3ft Al plate analysis:
    - FOS 2 (Yield) on bolt connections
    - Not within our desired requirements
    - Exploring other attachment alternatives
        - Moving bolts closer to center





Requirement #2: Lateral Load Resistance

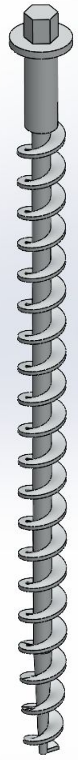
- Frame must hold against **730 lbf of horizontal force**
- Accounts for side-loading during **TVC, wind, or misalignment**
- FEM simulation ensures no failure or excessive deformation
- Test: Lateral load applied through dummy engine or thrust offset test

ANCHOR	LATERAL LOAD CAPACITY (lb)*
PE46	2,300
PE36	1,520
PE26	950
PE18	510
PE14	230
PE10	210

*In Class 2/3 soil

ETS Anchoring

1. The ETS will be anchored directly into soil with 26” Penetrators from American Earth Anchors.
2. This method was used for Elysium 1 and is proven successful.
3. Soil testing was done at site during Elysium 1 and the site was concluded to have a type 2 or 3 soil.
4. Penetrators have a respective pullout force of 3100 lb and 1100 lb for such a soil type and will hence allow the stand to withstand such forces



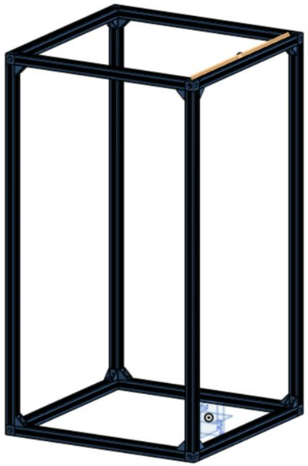
LOAD CAPACITY

Pullout strength with flight fully embedded

Soil Class 1	Soil Class 2	Soil Class 3	Soil Class 4	Soil Class 4
Hardpan Asphalt	Sandy gravel Very dense sand	Silty/clayey sand Silty gravel	Loose/med dense sands Loose sands Firm clays	Loose fine un-compacted sand
4,500 lb 20.0 kN	3,100 lb 13.8 kN	1,100 lb 4.89 kN	630 lb 2.80 kN	360 lb 1.60 kN

Extrusion and Connectors (Req 3 + 4)

1. The stand will use 45mm x 45mm aluminum extrusions for the external frame, components such as the platform for mounting the TCA and fluid tanks will be attached to this.
2. Corresponding 90 degree brackets will be used for connections.
3. Extrusions allow for a more modular design, they can be disassembled and reused for any future fires in any required orientation.
4. Structural integrity verified by Vention (4<FOS<5); in-house ANSYS verification in progress



Part Properties

In stock

90 Degree Inside Corner Bracket for 22.5 x 22.5 mm Extrusions

Order Details

Designs (1)

45mm Stand Design ME-OT-423708	x 1	**$1,480.61**
Shipping & Handling Calculated after shipping method		**TBD**
Subtotal		**$1,480.61**
BALANCE DUE		**$1,480.61 CAD**

Horseshoe (Req 1 + 4)

Custom **horseshoe-style base plates** are used to hold the aluminum extrusions in place at ground level.

The horseshoe design creates a **sleeve around the extrusion**, preventing lateral or rotational movement.

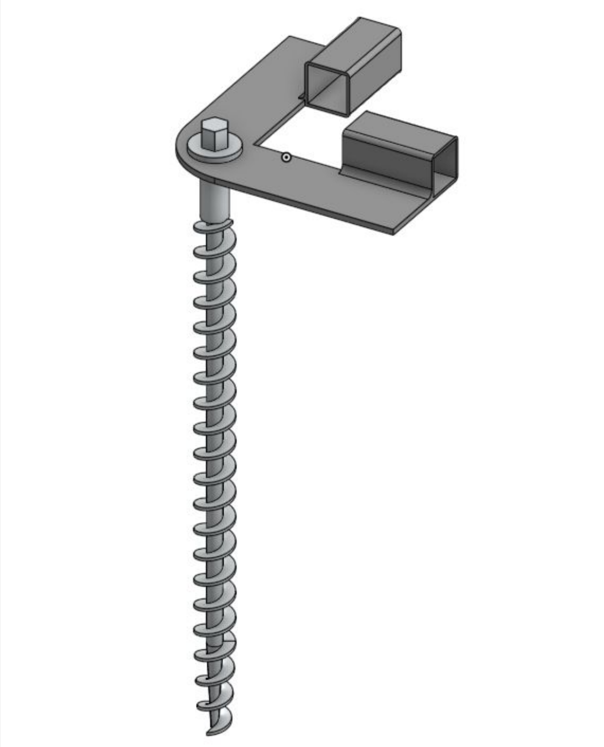
Each horseshoe is secured with **penetrator-style earth anchors** driven directly through the flat base and into soil.

This design allows **precise positioning, ground contact**, and strong anchoring while maintaining modularity for disassembly.

Cost Breakdown:

4ft Steel tubing - $55.42

Steel L - $100.68



~3’ Plate Analysis

A **3’ x 3’ x 0.5” plate of 6061 Aluminum** will serve as the interface between the thrust chamber and the vertical test stand.

The plate is **bolted directly onto the aluminum extrusion frame**, distributing load from the TCA and maintaining flatness under thrust.

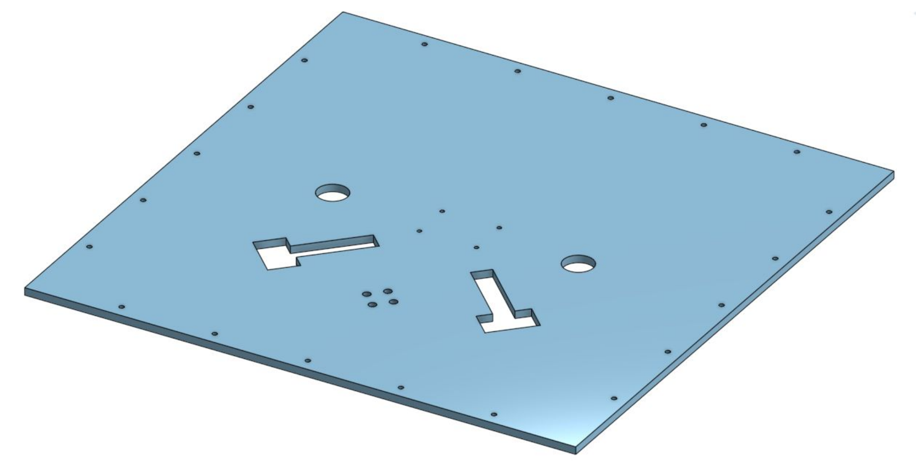
6061 Aluminum is chosen for its **high strength**

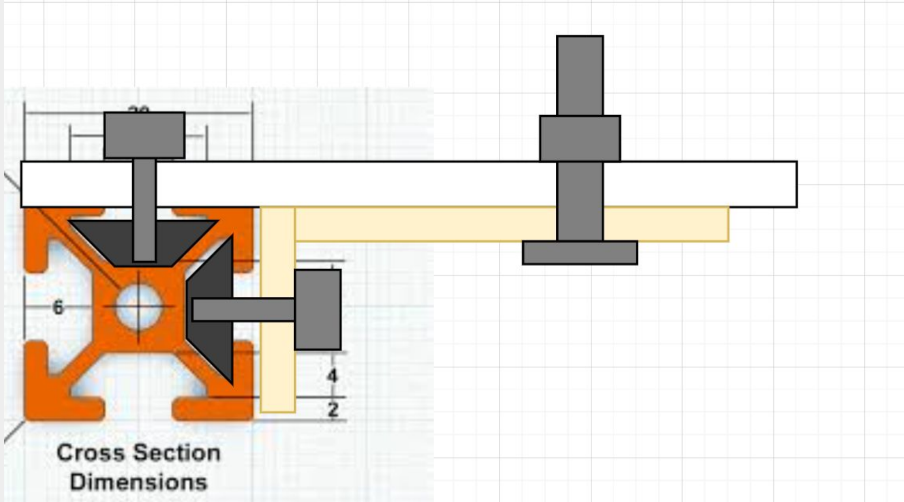
FEA was conducted to ensure the plate **withstands both static and dynamic loads** with a safety factor **<4**

Currently SF = 2.0 failing at bolts

-Upgrade in near future using brackets to reduce moment on bolts

Cost: The quote from our sponsor Send Cut Send came out to - $1592.33





Gimbal

- Gimbal Assembly
- Pins and Bolted Joints
- Deflection

Gimbal Requirements

#	Requirement	Rationale	Verification
6	The engine will be able to move 5 degrees in every direction while engine is firing	In order accurately control attitude and flight path in flight, the engine must be able to be gimbaled	Dry Run Test, Prototype

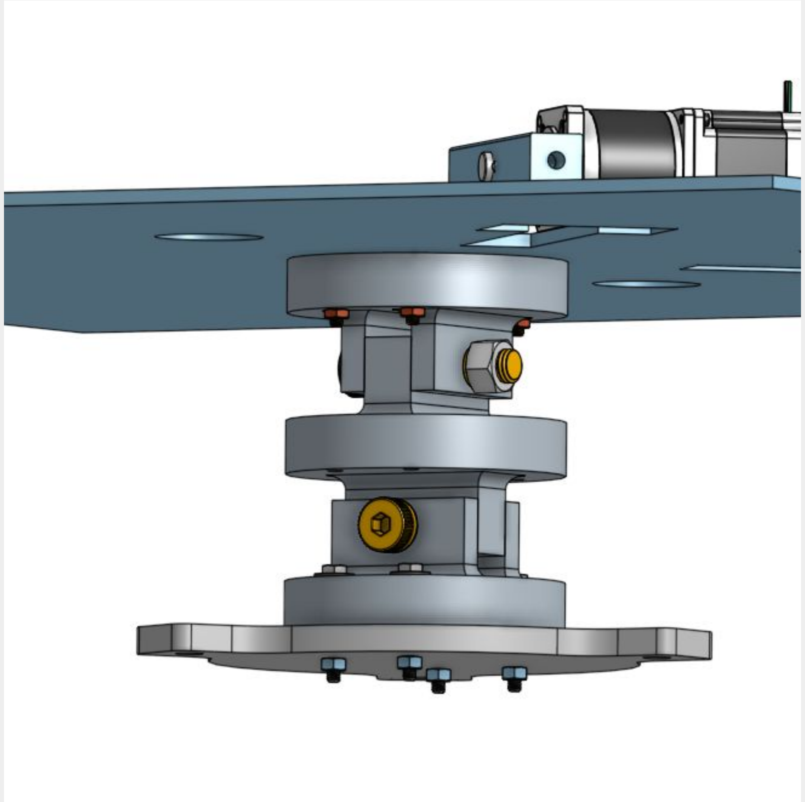
Assembly - Main Joint

Gimballing assembly includes:

- Machined Aluminum Gimbal Joints
- Alloy Steel Shoulder Screws
- Grade 8, 8” and 1-⅝”, bolts
- Washers
- High Strength Steel Nuts
- 304 Stainless Steel Interface Plate

Assembly will be done by hand on site using hand and power tools

Cost for assembly: $177.85



Assembly - Cost Breakdown

Item	Cost	Quantity	Acquiring From	Total Cost
Alloy Steel Shoulder Screw (1 count)	$28.82	3	McMaster-Carr	$86.46
2” ¼”-20 G8 Bolt (50 count)	$13.86	1	McMaster-Carr	$13.86
9” ¼”-20 G8 Bolt (1 count)	$8.09	5	McMaster-Carr	$40.45
High Strength Steel Hex Nut G8 ¼”-20 (100 count)	$7.21	1	McMaster-Carr	$7.21
High Strength Steel Hex Nut G8 ⅝”-11 (10 count)	$13.51	1	McMaster-Carr	$13.51
Zinc Flat Washer - ¼” (12 count)	$1.47	1	Home Depot	$1.47
Zinc Flat Stainless Steel Washer - ⅝” (4 count)	$1.47	3	Home Depot	$1.47

Design - Main Joint

Material: 6061 T6 Aluminum

Joint: Pin and Bolted

Interface Plates: Laser Cut 304 Stainless Steel

Contacts points will be properly lubed

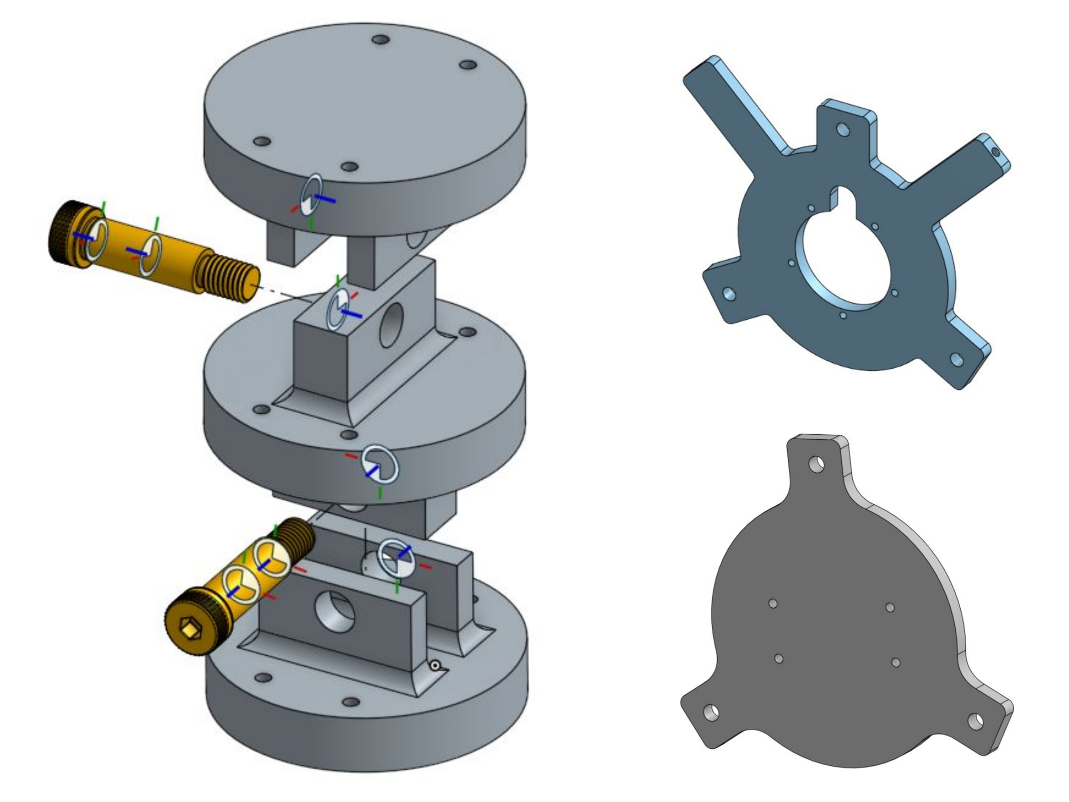
Loctite blue will be used to prevent our fasteners from loosening due to vibration

3-Piece machined aluminum gimbal allowing pitch and yaw control through the use of stepper motors

Machined in house using university resources (25$/hr)

Cost Breakdown:

- Gimbal Plate: $143.95
- Thrust Chamber Interface: $163.77
- Loctite Blue: $7.98

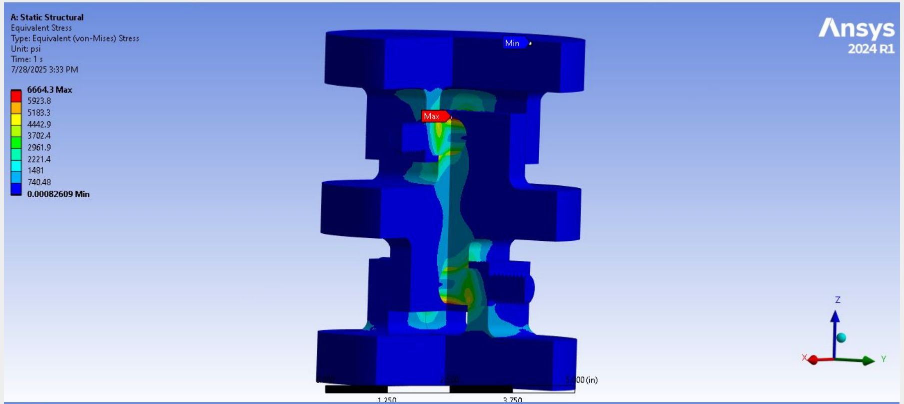


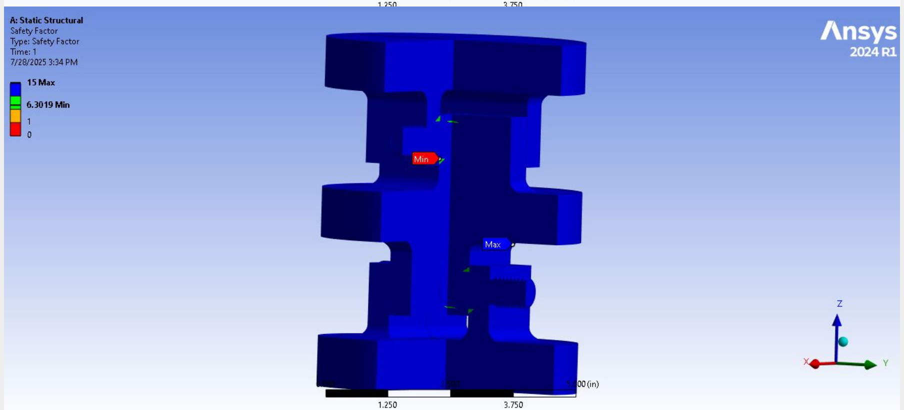
FEA Analysis through ANSYS (Static)

Conditions set for the simulation:

- Top gimbal face is set as fixed support to mimic the condition of the gimbal being bolted to the 3x3 plate
- Contacts are set to frictional with the coefficients set to .3 for aluminum-aluminum and .47 for aluminum-steel
- Standard Earth Gravity
- A force of 1500 lb though the bottom face of the bottom gimbal

Results: Observed a max stress of 6.6 ksi, significantly under our limit of 8ksi. A safety factor of 6.3 at the lowest point, well above our desired FOS of 4



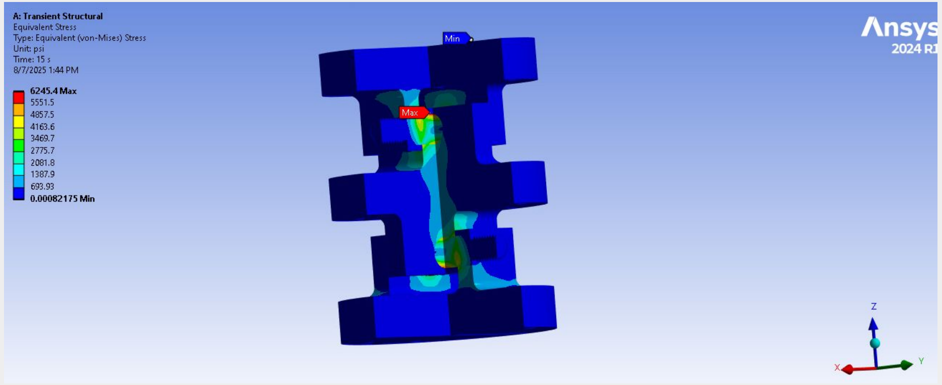


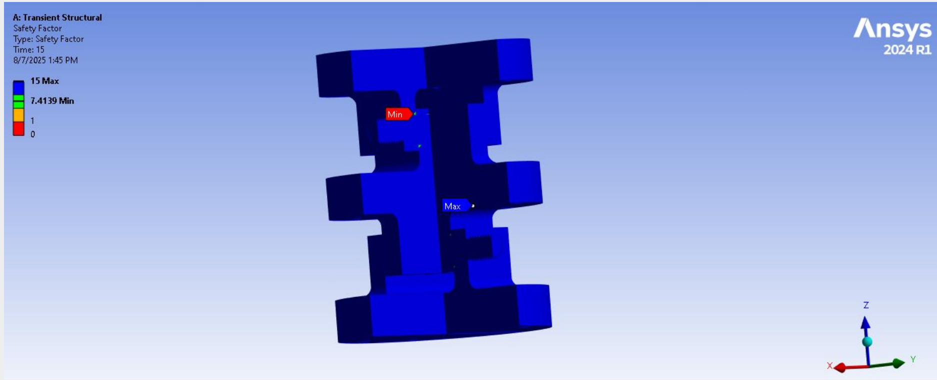
FEA Analysis through ANSYS (Transient)

Conditions set for the simulation:

- Top gimbal face is set as fixed support to mimic the condition of the gimbal being bolted to the 3x3 plate
- Contacts are set to frictional with the coefficients set to .3 for aluminum-aluminum and .47 for aluminum-steel
- Standard Earth Gravity
- A force of 1500 lb though the bottom face of the bottom gimbal
- This sim was done over 15 secs with steps every second

Results: Observed a max stress of 6.2 ksi, significantly under our limit of 8ksi. A safety factor of 7.4 at the lowest point, well above our desired FOS of 4. These results are consistent with the static simulation



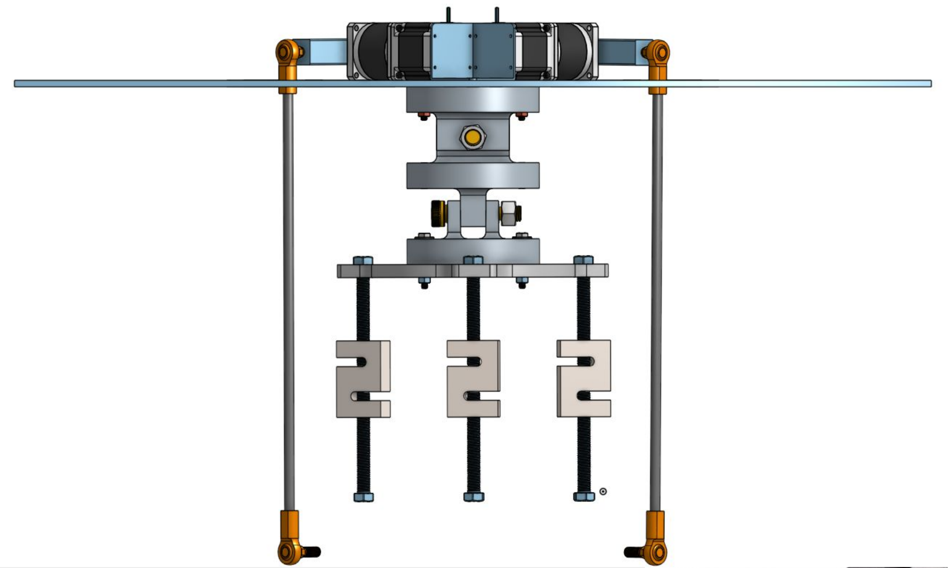


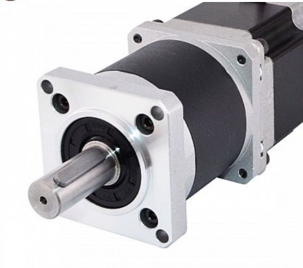
Assembly - Actuation

Actuation assembly includes:

- x2 Nema 23 Stepper Motor ~$184
- x2 6”x1”x1” Al 6061-T6 Square Tubing ~$10
- x2 14 mm 1215 C Steel Shaft Collars ~$16
- x1 Aluminum 6061-T6 Bended Stepper Corner Brace ~$35
- x4 ⅜”-24 steel ball joints ~$40
- x2 ⅜”-24 high strength steel threaded rods ~$69
- Total Cost: ~$371

Assembly will be done by hand and on site.





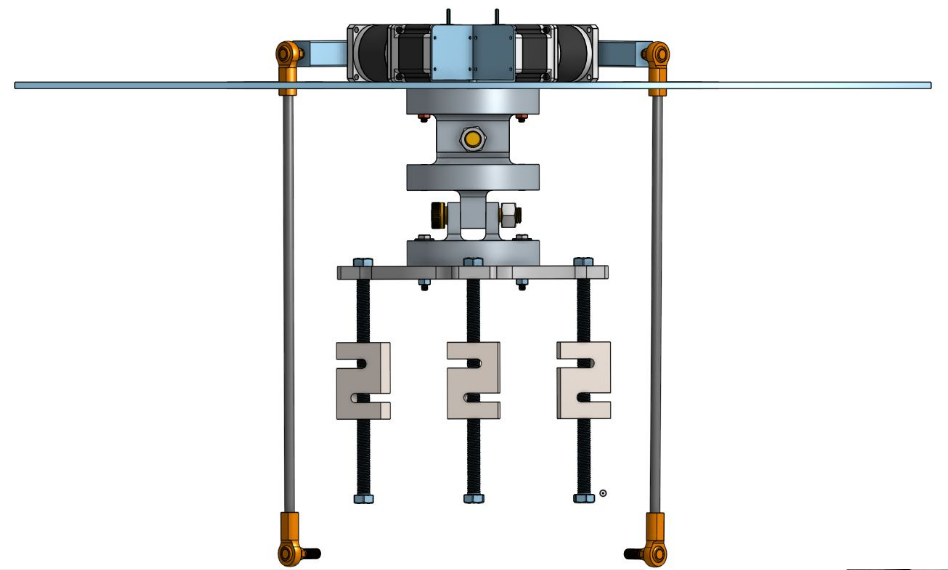
Design - Actuation

Material: Steel and Aluminum

Steppers (Keyed): Seated 90 degrees apart from via a corner brace connected to 3ft base plate (fixed support). The steppers keyed shafts are connected to a machined aluminum keyed interface with shaft collars at the end

Joints: Each arm assembly has 2 ball joints, upper joint connects machined aluminum (horizontal arm) to the threaded rod (vertical arm). The bottom joint connects the threaded rod to the interface plate.

Gimballing: Gimballing will be achieved by commanding a stepper motor to an angle to get a desired angle out. This will be figured out by the following math (courtesy of Devin Smith - AVI Member)

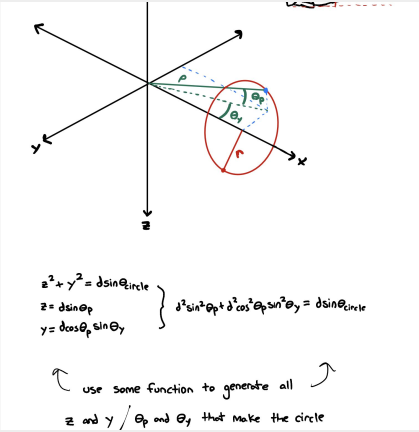


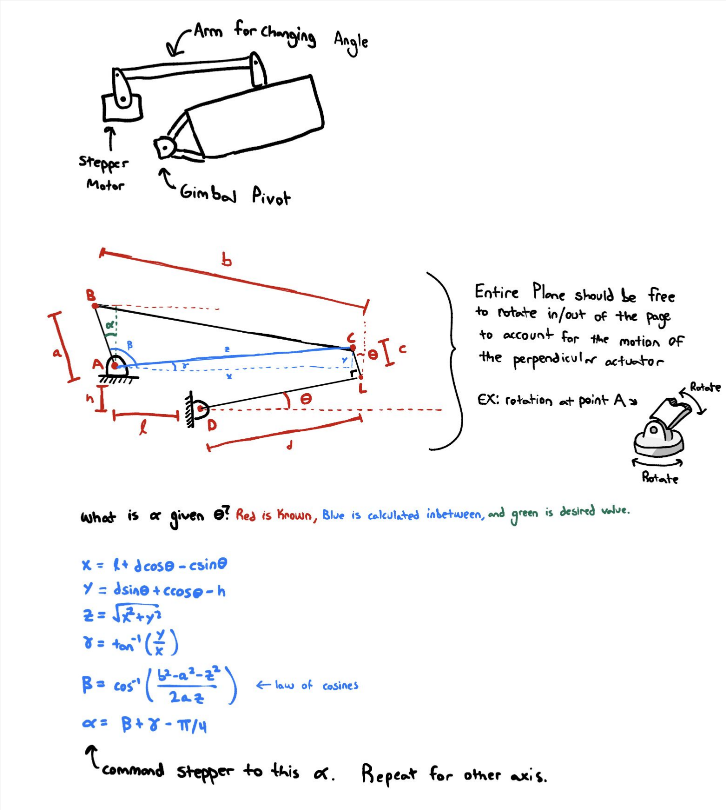
Math - Actuation

Math behind gimbaling - Courtesy of Devin Smith - AVI Member.

More info during AVI CDR

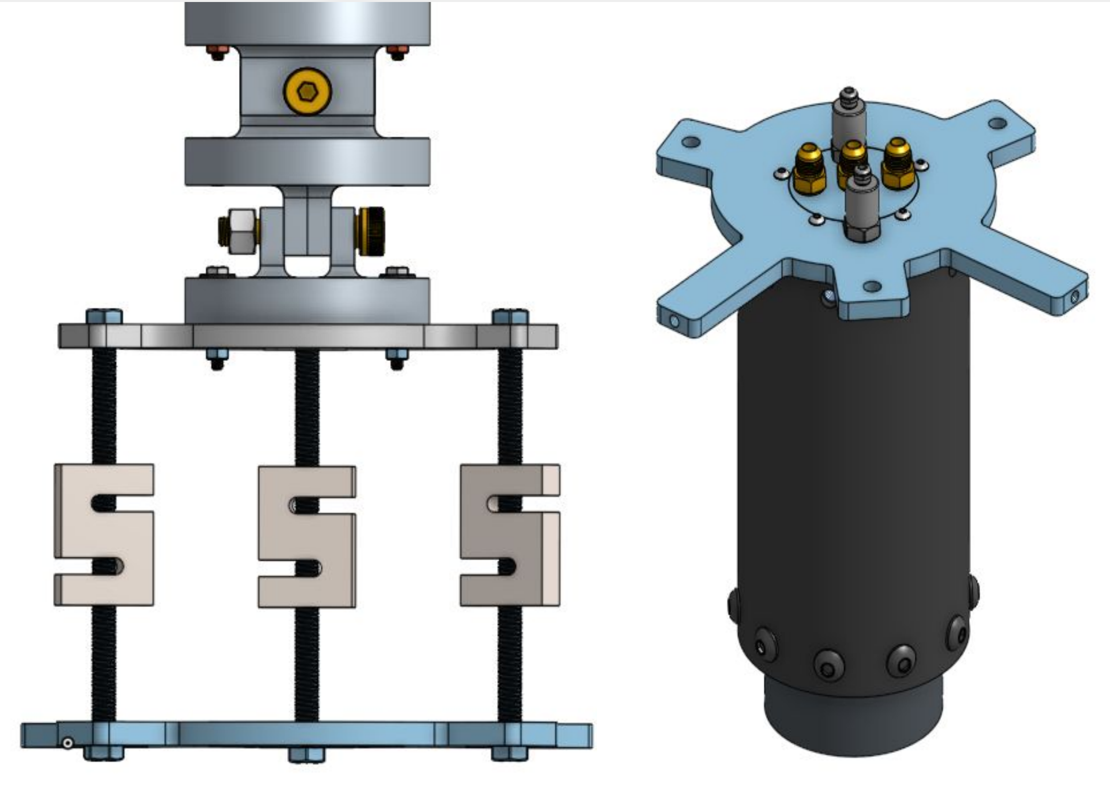
Planned Testing: Preliminary testing of steppers w/ fixed support + arms + interface plate (gimbal movement). Then during a dry run with majority of full assembly (full weight).





Interfacing with Thrust Chamber Assembly

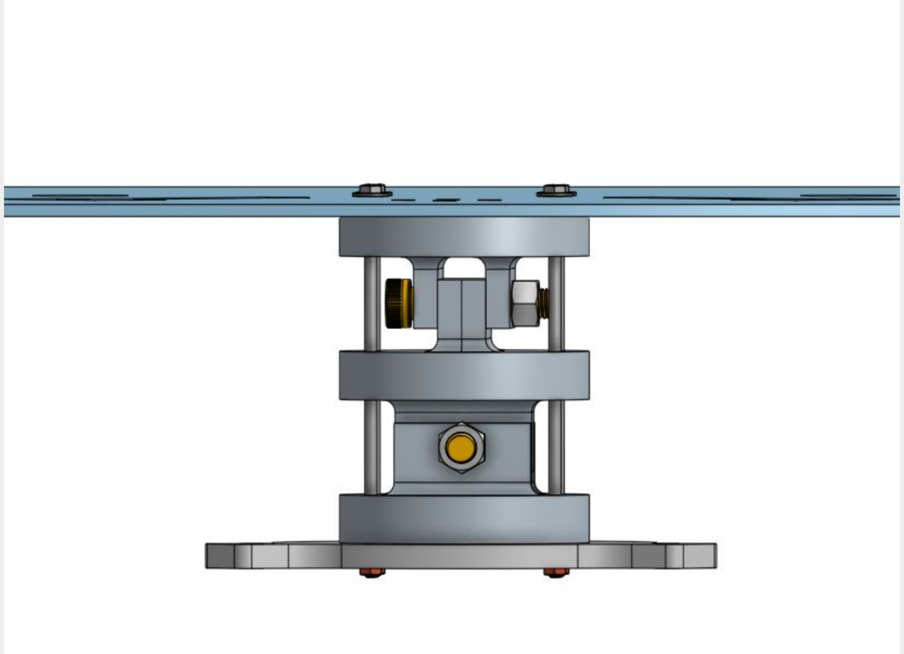
Using a similar set up as Elysium 1, the Thrust Chamber Assembly will be attached to the Gimbal through threaded rods connected to the two interface plates via S-type load cells and bolted joints.



Locking the Gimbal

Not all fires will feature the use of Thrust Vector Control, however the gimbal will still be present. Therefore precautions must be taken to prevent any unwanted pitch or yaw deflections.

Solution: Using 8” grade 8 bolts that slot through the gimbal in its entirety will prevent longitudinal and lateral movement



Wooden Stand

- Complete Assembly
- FOx Stand
- Table Support
- Load Cell Mechanism
- Wooden Stand Documentation

Wooden Stand Requirements

#	Requirements	Rationale	Verification
8.1	Wooden Stand must support FOx tank as well as other components such as pipes and firmware.	Wooden Stand must be able to support these functions for hotfires, logging data, and ensuring accessibility for maintenance.	Part and Assembly prints, discussion with leads, and observation.
8.2	Wooden Stand must support load from components as well as withstand external forces such as vibrations and wind.	Wooden Stand must be able to withstand these forces to support the components along with safety.	Perform initial calculations, FEA, observations.
8.3	User accessibility.	Several components will be dependent on the Wooden Stand. To perform proper maintenance, it is important to allow for accessibility.	Part and Assembly Drawings.

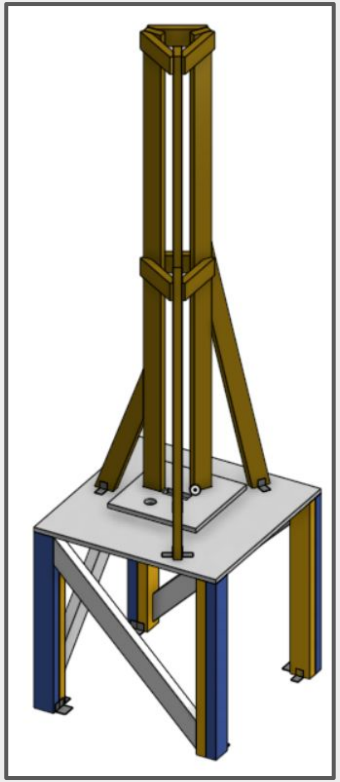
Complete Assembly

Total stand must support 500 lb given a FoS 4. This means that the stand must support a static load from the FOx tank as well as the piping and firmware. Load will primarily be on the Table Support with the FOx stand acting as a method of securement for FOx tank. Methods of load bearing will consist of:

- Metal Brackets and bolts/screws for securement between wooden pieces.
- Metallic cables for ground securement.
- Vertical & Cross-sectional wooden support for load bearing and rigidity.
- Set up will be secured via Bolts to 3’ x 3’ x ½” 304 Stainless steel plate.

Results:

Given initial calculations, the current stand should be able to withstand the loads predicted during hotfires.



FOx Stand

Part List:

2"x4"x8' Normal Wood Plank	0	https://www.homedepot.com/p/WeatherShield-2-in-x-4-in-x-6-ft-2-Prime-Pine-Ground-Contact-Pressure-Treated-Southern-Yellow-Pine-Lumber-253920/206967815	$5.88	$0.00
2"x4"x8' Nominal Wood Plank	7	https://www.homedepot.com/p/WeatherShield-2-in-x-4-in-x-8-ft-2-Prime-Ground-Contact-Pressure-Treated-Lumber-253920/206967811	$5.58	$39.06
2"x4"x6' Nominal	7	https://www.homedepot.com/p/2-in-x-4-in-x-8-ft-SYP-2-Standard-Grade-Dimensional-Lumber-1020/305625813	$3.85	$26.95
4'x8' Plywood	1	https://www.homedepot.com/p/OSB-7-16-Application-as-4ft-X-8-ft-Sheathing-Panel-386081/202106230	$15.28	$15.28
2"x1.5'" Simpson Strong-Tie	22	https://www.homedepot.com/p/Simpson-Strong-Tie-2-in-x-1-1-2-in-x-1-3-8-in-Galvanized-Angle-A21/100374951	$0.75	$16.50
#8x1-0.5" Wood Screws	1	https://www.homedepot.com/p/Everbilt-8-x-1-1-2-in-Zinc-Plated-Phillips-Flat-Head-Wood-Screw-100-Pack-801842/204275487	$11.47	$11.47
#12x2" Phillips Flat Head	1	https://www.homedepot.com/p/Everbilt-12-x-2-in-Phillips-Flat-Head-Zinc-Plated-Wood-Screw-50-Pack-801982/204275510	$10.32	$10.32
2"x1.5"x2.75" Simpson Strong-Tie	4	https://www.homedepot.com/p/Simpson-Strong-Tie-2-in-x-1-1-2-in-x-2-3-4-in-Galvanized-Angle-A23/100374944?MERCH=REC-_-pipsem-_-100374951-_-1-_-n/a-_-n/a-_-n/a-_-n/a-_-n/a	$0.98	$3.92
			Sum	$123.50

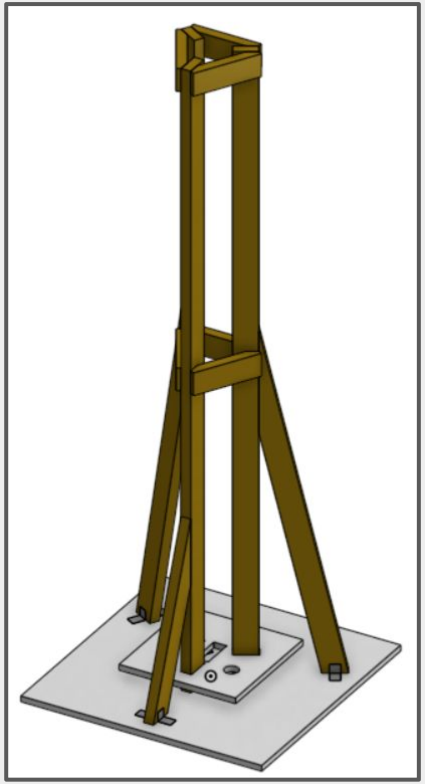
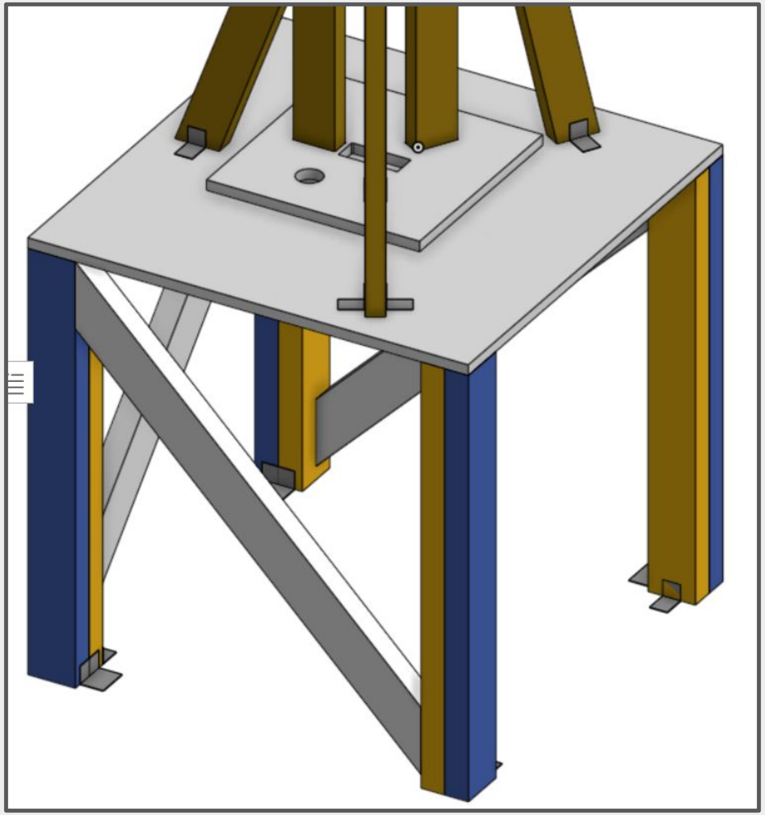


Table Support

Part List:

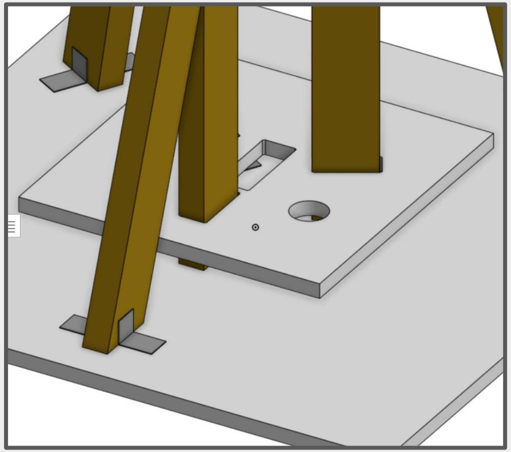
2"x4"x6' Nominal	7	https://www.homedepot.com/p/2-in-x-4-in-x-8-ft-SYP-2-Standard-Grade-Dimensional-Lumber-1020/305625813	$3.85	$26.95
2"x1.5"' Simpson Strong-Tie	8	https://www.homedepot.com/p/Simpson-Strong-Tie-2-in-x-1-1-2-in-x-1-3-8-in-Galvanized-Angle-A21/100374951	$0.75	$6.00
2"x1.5"x2.75" Simpson Strong-Tie	8	https://www.homedepot.com/p/Simpson-Strong-Tie-2-in-x-1-1-2-in-x-2-3-4-in-Galvanized-Angle-A23/100374944?MERCH=REC-_-pipsem-_-100374951-_-1-_-n/a-_-n/a-_-n/a-_-n/a-_-n/a	$0.98	$7.84
2 in. x 4 in. x 3 ft. #2 Premium Grade Dimensional Lumber	8	https://www.homedepot.com/p/ProWood-2-in-x-4-in-x-3-ft-2-Premium-Grade-Dimensional-Lumber-271736/300524956	2.25	$18
		Sum		$58.79

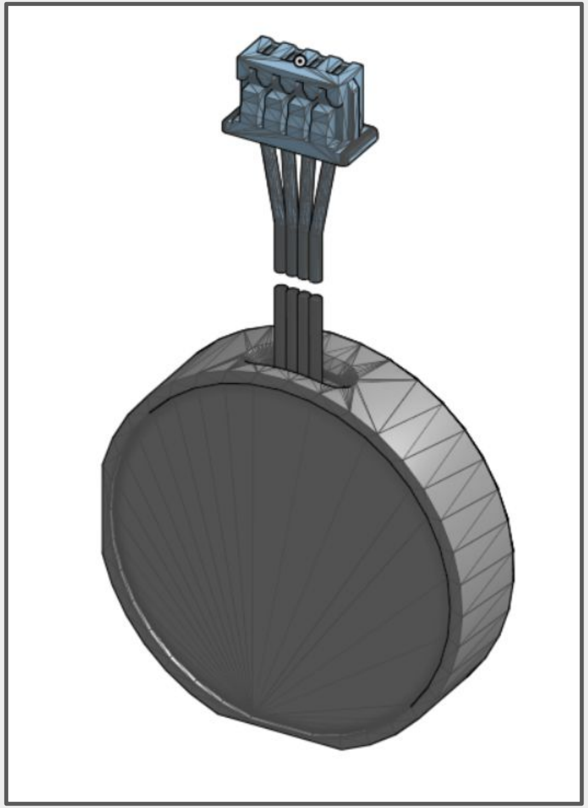


Load Cell Mechanism

Load Cells used are the c-20009605-52-d-3d. We will use two of these and will be placed beneath the 1’ x 1’ wooden plate to study the rate of change for the mass of the FOx tank.

Horizontal Restriction will consist of wooden dowel pins that will be drilled into the 3’ x 3’ wooden plate. Total cost for this is $3.28





Wooden Stand Documentation / Action Items

Some Action Items for the Wooden Stand consist of:

- Triple Checking BOM
- Creating accurate prints

Recommendations for future adjustment:

- Install Eye bolts with cables on corners to reduce vibration chances.
- Consider securing top of FOx stand to ground for better security.

Flame Diverter

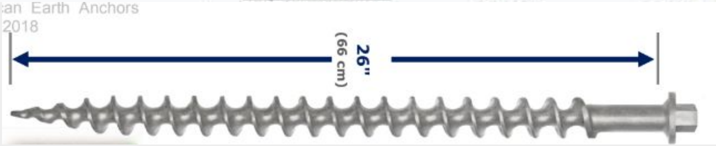
- Steel Frame
- Penetrator
- J plate
- Flat plate

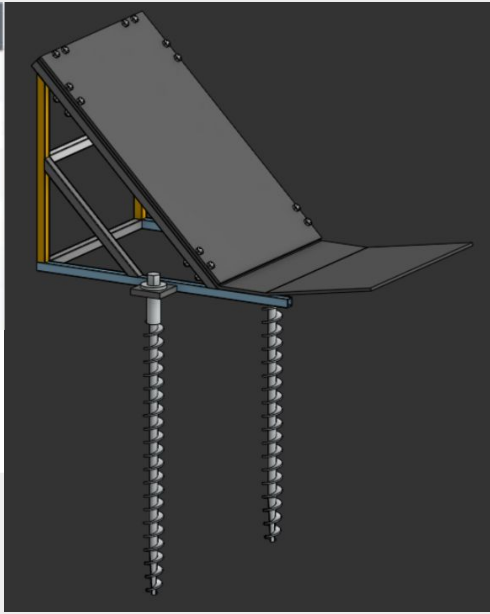
Requirement #2 : Flame Diverter Withstands Thrust

#	Requirement	Rationale	Verification
2.1	Penetrators, joints and connections must remain secure	Entire flame diverter must remain in place and static during the fire. Bolts secured properly will ensure structural integrity of assembly.	Inspect assembly pre and post test
2.2	The entire assembly (Steel frame, Penetrator, J plate, Flat Plate) must handle forces slightly over 1,500 lbs	Ensures structural integrity for the assembly (Steel frame, Penetrator, J plate, Flat Plate) under peak thrust of 1,500 lbs force for 15 seconds.	ANSYS FEA
2.3	Structure is tested with minimum safety factor of 4.0 on expected peak loads	Accounts for uncertainties in material behavior	ANSYS FEA

Assembly

Item	Quantity	Link	Cost	Total Cost
SendCutSend Diverter Plates (304)	1	https://cart.sen	$727.48	$727.48
Carbon Steel 3/4" 6ft Telescoping Weld-Together Rails (1471T51)	3	https://www.mc	$21.58	$64.74
Hillman-6-in-x-18-in-Cold-Rolled-Steel 16 Gauge	1	https://www.lov	$10.48	$10.48
3/8"-16 Lg. 1 3/4" Partially Threaded Zinc-Plated Medium-Strength Grade 5 Steel	1	https://www.mc	$10.45	$10.45
3/8"-16 Corrosion-Resistant Stainless Steel Nuts (92673A125) (25 per pkg)	1	https://www.mc	$2.55	$2.55
SendCutSend Penetrator Plate	1	https://cart.sen	$38.82	$38.82





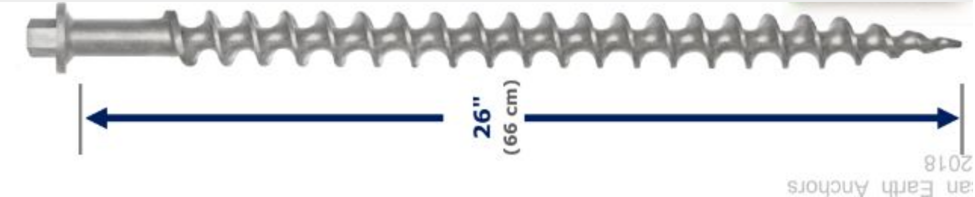
Anchoring:
26” Penetrators from American Earth Anchors

Steel Plates:
Top plate - $290.32
Bottom plate - $456.30

Requirement 2.1 : Secure penetrators, connections and joints

Solution: Use 26” Penetrators from American Earth Anchors to ensure stand is secured into the soil and ASTM Grade B8 18-8 Stainless Steel Bolts that are 1 ¾” for bolted plate to test stand. Welding for structural components and non-removable hardware.

Verification: Pre and post test inspection for structural integrity to ensure components are attached and secured. Perform vibration testing simulating the expected frequency range during engine fire.



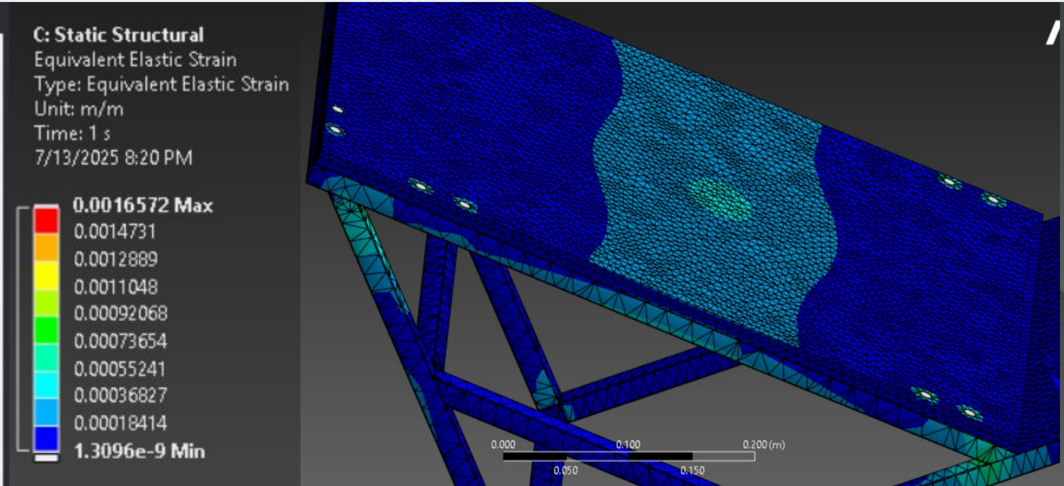
ANCHOR	LATERAL LOAD CAPACITY (lb)*
PE46	2,300
PE36	1,520
PE26	950
PE18	510
PE14	230
PE10	210

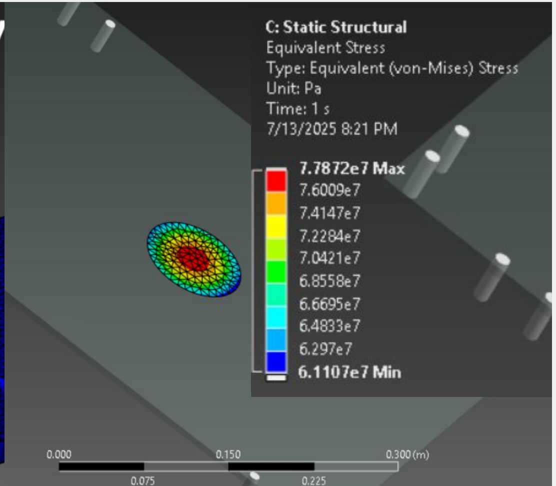
*In Class 2/3 soil

Requirement 2.2 : Entire assembly handles >1,500 lbf thrust

Solution: Steel frame, Penetrators, J plate, Flat Plate are designed to handle over 1,500 pounds.

Verification: FEA confirms that all components withstand the required load without failure. Ansys run at 1,800 lbf shown below.

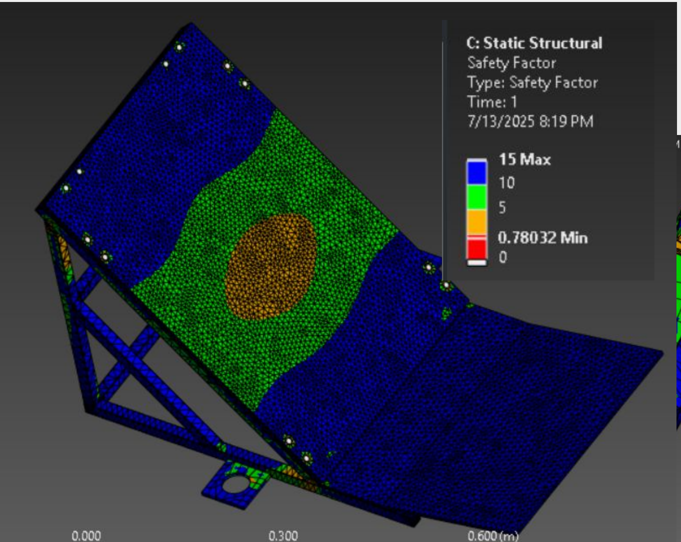


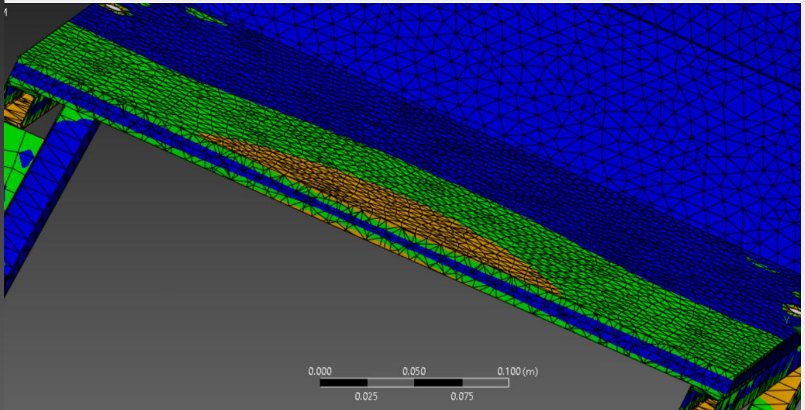


Requirement 2.3 : Flame diverter tested with minimum FOS of 4.0

Solution: Minimum FOS of 4 ensures structure can handle uncertainties in load distribution.

Verification: FEA demonstrates that the FOS ranges from around .78 to 15. Top plate may be replaced if material fails. Rest of assembly demonstrates non-critically loaded areas.

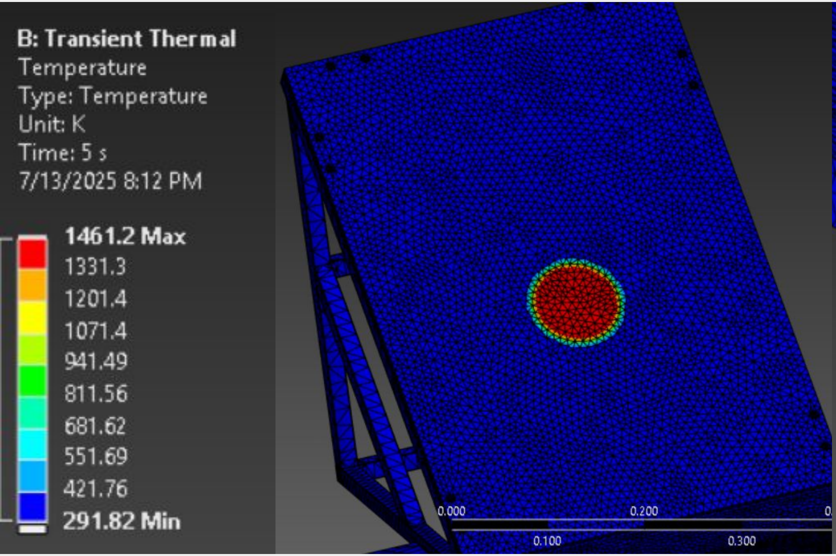


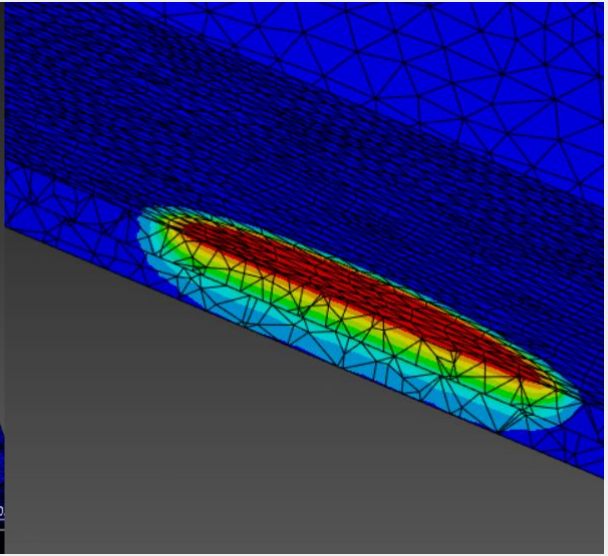


Requirement 5 : Entire assembly withstands thermal load

Solution: Top and bottom plate are designed to handle 1,500K for 15 seconds

Verification: FEA confirms that all components withstand the required thermal load. Ansys run at 10,000 w/m^2 at 1,500 lbf for 5 seconds.





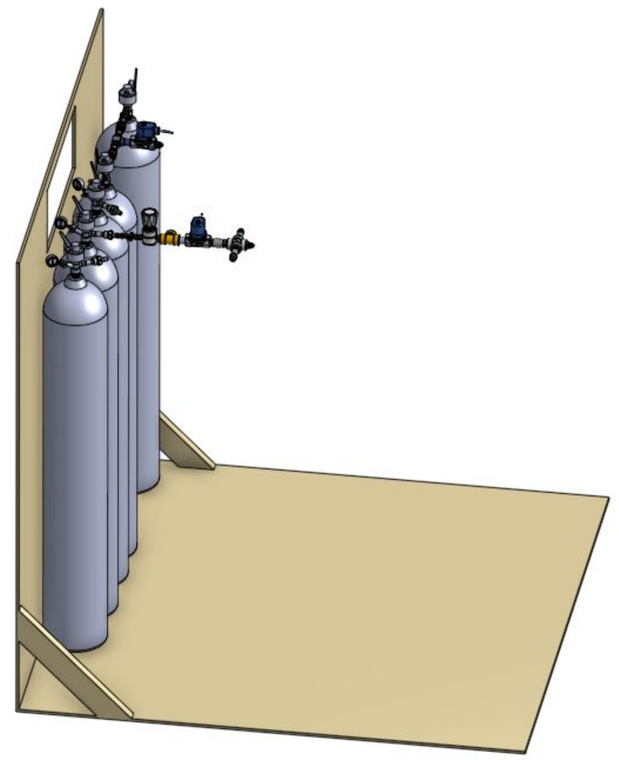
Bottle Stand

- Flat board
- Vertical board
- Screws and brackets

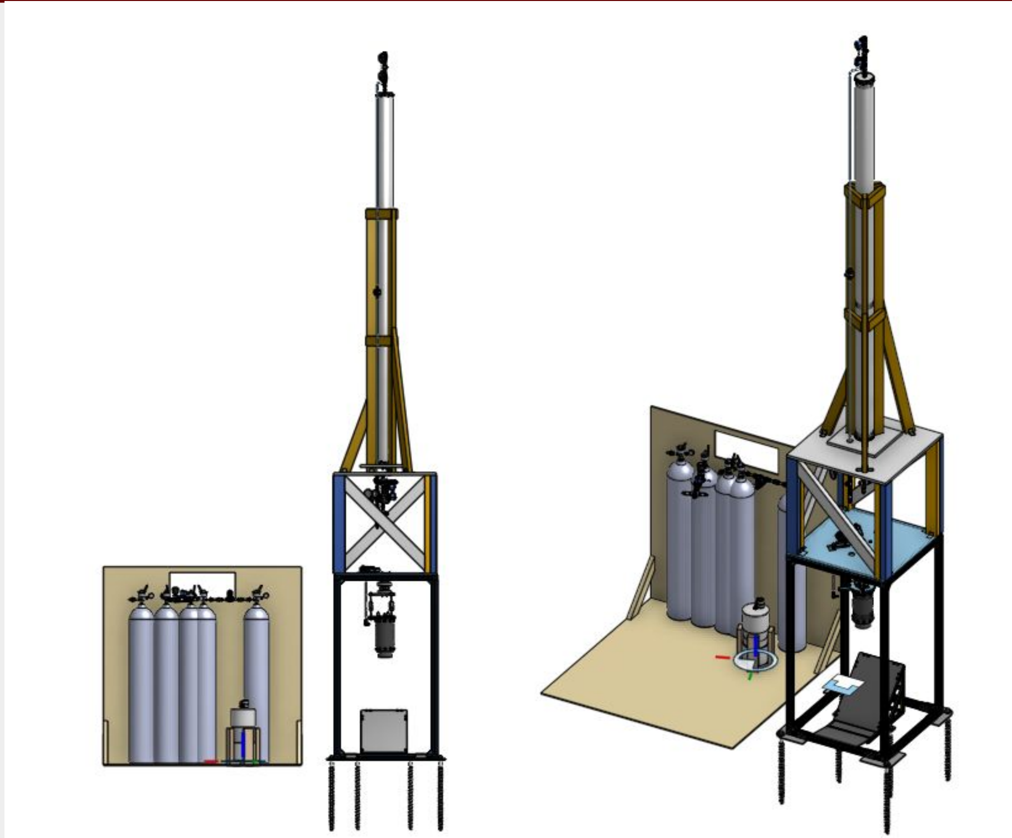
Bottle Stand

Due to the presence of two class 2 (N2O & GN2) substances and a single class 3 substance (Ethanol) extra precautions are being taken to ensure operator safety.

A bottle stand will be constructed to provide a stable platform for the K-bottles to be secured onto using ratchet straps, as well as a barrier between the operator and the bottles



Full Assembly



Overview of Assembly Procedure - Structures Focused

1. Aluminum stand first w/ ~3 ft plate
   a. Assembled on side, then raised up
   b. 3 foot plate goes on when vertical, to ensure bolt holes are lined up properly
2. Main engine assembly
   a. Gimbal 3 piece assembled and attached, locking bolts temporarily attached (to prevent movement during rest of assembly)
   b. Gimbal interface plate attached
   c. Loaded cells + threaded rods attached to gimbal interface
   d. TCA interface plate + TCA attached
3. Actuation arms assembly installed
4. Flame diverter assembly installed
5. FOx tank stand assembly installed
6. Bottle stand assembled for K-Bottles
7. (After FSYS install complete) Gimbal locking bolts removed

Future Plans and Total Cost

- Continued analysis on structural components
- Begin machining of gimbal components
- Increasing connections safety factor using using L-shaped bracket reinforcements
- Shorten Wood Table

Penetrators and Gimbal Plates

- Sponsored

Aluminum Stand

- $1,100

Aluminum Plate

- $1,600

Aluminum Raw Stock

- Already Bought

Wood and Hardware

- $200

Gimbal Hardware

- $175

Total Expected Future Cost: $**3,175**

Questions

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